

So,

$$\frac{\partial \mathcal{L}_{CE}}{\partial z} = (a-y) \quad \text{and} \quad \frac{\partial \mathcal{L}_{MSE}}{\partial z} = 2(a-y) \cdot a(1-a)$$

Notice that the cost for MSE has an extra factor of $2a(1-a) = 2a - 2a^2$

Taking $\frac{d}{da} 2(a-a^2) = 2(1-2a)$, we find the max value of this function to be at $a = \frac{1}{2}$

$$(1-2a=0 \rightarrow a = \frac{1}{2})$$

$$\text{So, } f\left(\frac{1}{2}\right) = 1 - \frac{1}{2} = \frac{1}{2}$$

Therefore $\forall a (2a(1-a) \leq \frac{1}{2} \mid 0 < a < 1)$.

Thus to achieve a comparable rate of gradient descent (at best with $a \approx \frac{1}{2}$):

$$2 \cdot lr_{CE} \gtrsim lr_{MSE}.$$

Since values of a fluctuate around $0 < a < 1$, this factor of 2 is a best-case scenario, hence why our optimal learning rate for this model is 10x-100x larger than it would be using cross-entropy loss.